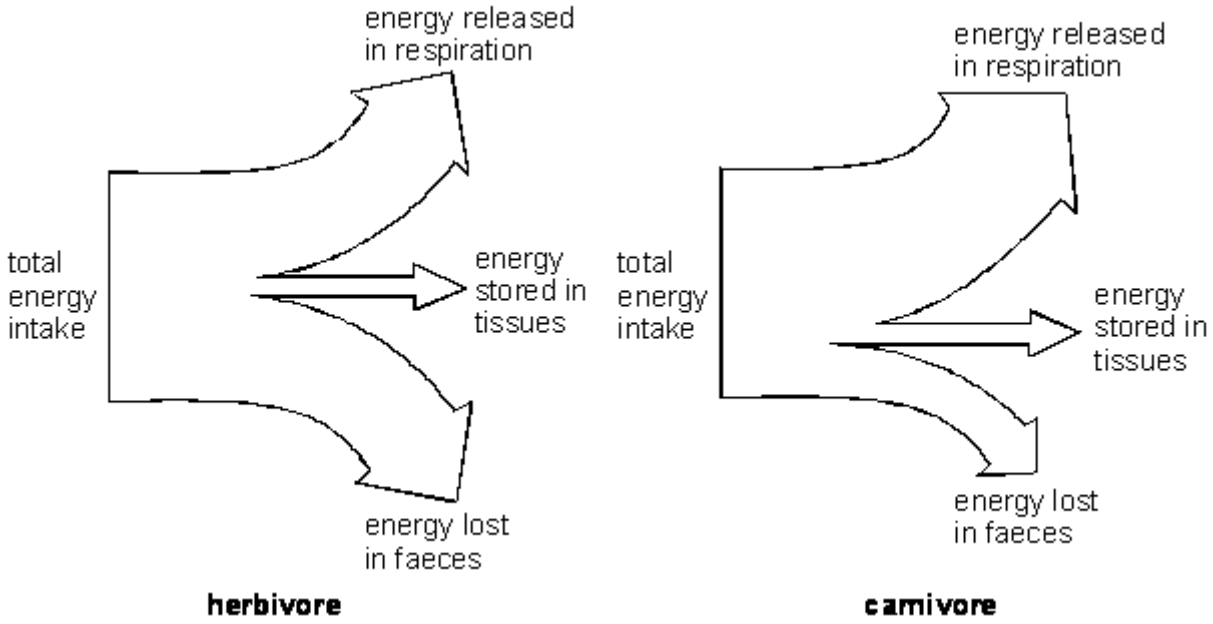
Q1. The diagrams below represent what happens to the energy in the food eaten by a herbivore and a carnivore.

The **width of each pathway** indicates the amount of energy gained or used in a particular way.



(a) (i) What percentage of the total energy, taken in by a herbivore, is stored in its tissues? Use the diagram to help you answer.

..... %

1 mark

(ii) The energy stored in an animal's tissues is passed on to the next animal in the food chain.
Use information in the diagrams above to explain why there are usually no more than four or five stages in a food chain.

.....

.....

1 mark

(b) Respiration takes place in cells, in structures called mitochondria. Why do muscle cells contain large numbers of mitochondria?

.....

.....

1 mark

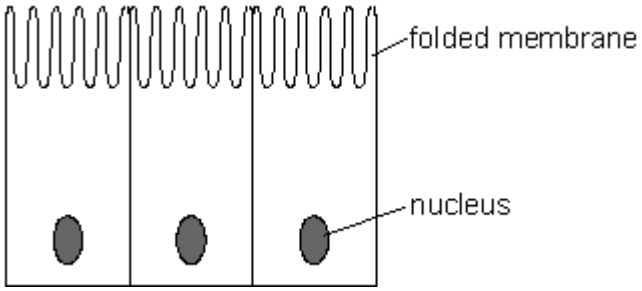
(c) Cows eat plants, but **cannot** digest the cellulose cell walls.
Micro-organisms in the cow's stomach are able to digest the plant cell walls.
Suggest why cows **cannot** digest the cell walls but micro-organisms can.

.....

.....

1 mark

(d) The diagram below shows cells from the inner lining of a mammal's intestine.



The cell membranes in contact with the food are folded.
Explain why it is an advantage that these cells are adapted in this way.

.....

.....

.....

2 marks
Maximum 6 marks

Q1.

(a) (i) 10

accept answers from 7 to 13

1

(ii) any **one** from

- only some **or** 10% of the energy intake is passed on to the next level in the food chain
- only some **or** 10% of the energy is stored in tissues
- some energy is wasted at each stage
- less energy is passed on to the carnivore

consequential marking applies

accept the percentage given in part (a) (i)

1

(b) a lot of energy is needed for muscle contraction

1

(c) any **one** from

- a cow cannot produce the correct enzyme
- micro-organisms produce the correct enzyme **or** cellulase

1

(d) folds increase the surface area

1

any **one** from

- more absorption takes place
- absorption takes place more efficiently
- absorption takes place more quickly

1

[6]